A Appendix

A.1 GLOW Algorithm

Algorithm 1 GETPROMPT: Generate GLOW Prompt from Input OWA Question over KG

Require: Q the user input question, $GLOW_v$ The system variation .

function GETPROMPT(Q,$GLOW_v$)
- $Q_n, Q_e, KG \leftarrow$ entityExtraction(Q) ▷ extract the question's node and edge
- $KG_{Sc} \leftarrow$ getKGSchema(KG) ▷ Load the KG schema triples
- $v_t, e_t \leftarrow$ ER-Linking(Q, KG_{Sc}, Q_n, Q_e) ▷ Link the question's node and edge crossponding KG's node and edge URIs
- $\mathcal{RC}_q \leftarrow$ TextToSPARQL(KG_{Sc}, v_t) ▷ generate $\mathcal{RC}$ equivalent SPARQL query
- $\mathcal{RC}_{Triples} \leftarrow$ execSPARQL($\mathcal{RC}_q, KG$) ▷ Execute the $\mathcal{RC}$ SPARQL Query
- $L \leftarrow$ getPossibleLabels(v_t, e_t) ▷ Extract the set of possible labels
- $RC \leftarrow$ RC-Serialization($\mathcal{RC}_{Triples}$) ▷ Serialize the $\mathcal{RC}$ triples into text.
- $GNN_{Ans} \leftarrow$ GNNPredict(v_t, e_t) ▷ predict the top-K GNN answers for v_t and e_t
- $P \leftarrow$ getPrompt($Q, v_t, e_t, L, \mathcal{RC}, GNN_{Ans}, GLOW_v$) ▷ generate the GLOW's variation prompt
- **return** P

end function

A.2 GLOW Prompt Examples

User Prompt:

Predict the chemical kingdom for the drug Yohimbine from the BioKG knowledge graph.
Answer:

Question Understanding Prompt:

<system>: You are an an expert Entity-Extraction NLP system.
<user>: Given the following question, identify 1-The question main entity type, 2- the main entity, 3- the prediction label and 4- the KG name.
Question: { }
Answer:
1-Question main entity: 2-Main Entity:
3-Prediction label: 4-KG name:

Entity/Relation Linking Prompt:

<system>: You are an an expert knowledge graph entity-relation Linking NLP system.
<user>: Given the following KG Schema in the basic graph pattern CSV format, one predicate per line: node type, relation, node type.

KG Schema:
{ }

1- What is the node type in the schema that corresponds to {main entity}? Return only the name.
2- Choose from the schema the BGP (node type, relation, node type) that describes the {main entity type} value {main entity}. Return only the BGP.
3- Choose from the schema the BGP (node type, relation, node type) that describe the {main entity type } {prediction label}. Return only the BGP.

Answer:
1-
2-
3-

Question To SPARQL Prompt:

<system>: You are an an expert text-To-SPARQL translation system.
<user>: Given the following KG Schema in the basic graph pattern CSV format, one predicate per line: node type, relation, node type.

———

KG Schema:
{}

———

Write a SPARQL query that selects the {main entity type} that satisfy the following BGPs.
1- {question node type} ,[label/name/titile] ,{question node}
2- {Other BGPs}
graph prefix: {KG Prefix}

————

Answer: SPARQL Query
Do not return any explanation or reasoning details.

SPARQL Query Example:

```
PREFIX biokg: <http://www.biokg.com/>
SELECT ?drug as ?vt ?kingdom as ?vl
WHERE {
VALUES ?name { "Yohimbine" }
?drug biokg:NAME ?name .
?drug biokg:KINGDOM ?kingdom .}
```

Basic Prompt:

<system>: You are an expert open world question answer system.
<user>: Predict the {**Prediction Label**} for the following {**question node type**} from {**KG details**} knowledge graph.
- Do not return any context or analysis.
- Help: The possible list of {**Prediction Label Type**}s are: [{ **Labels List**}]
- Question: {**question node**}

Answer

To generate an instance of this template, replace the question node type, KG, and prediction label with values from one of the queries in table 5. **An example prompt for OWA-Q #1**:

<system>: You are an expert open world question answer system.
<user>: Predict the **Kingdom** for the following **Drug** from the **BioKG , a Biomedical** knowledge graph.
- Do not return any context or analysis.
- Help: The list of **Kingdoms** are: [Organic,Non-Organic]
- Question: **Durg:** Yohimbine

Answer:

GLOW-G Prompt:

<system>: You are an expert open world question answer system.
<user>: Predict the **Kingdom** for the following **Drug** from the **BioKG , a Biomedical** knowledge graph.
- Do not return any context or analysis.
- Help: The possible list of **Kingdoms** are: [**Organic,Non-Organic**]
- Question: **Durg:** and its associated triples. Capsaicin: [("DDI", "DB13677"), ..]

Answer:

GLOW-N Prompt:

<system>: You are an expert open world question answer system.
<user>: Predict the **Kingdom** for the following **Drug** from the **BioKG , a Biomedical** knowledge graph.
- Do not return any context or analysis.
- Help: The possible list of **Kingdoms** are: [**Organic,Non-Organic**]
- Verify the following list of GNN Answers: [Organic,Non-Organic, ..]
- Question: **Durg:** Yohimbine
Answer:

GLOW-GN Prompt:

<system>: You are an expert open world question answer system.
<user>: Predict the **Kingdom** for the following **Drug** from the **BioKG , a Biomedical** knowledge graph.
- Do not return any context or analysis.
- Help: The possible list of **Kingdoms** are : [**Organic,Non-Organic**]
- Verify the following GNN Answer: [Organic]
- Question: **Durg** and its associated triples. Capsaicin: [("DDI", "DB13677"), ..]
Answer:

A.3 LLM as-a-Judge Prompt

<system>: You are an expert LLM-as-a-Judge system.
<user>: Given the following list of predicted and true pairs of values.
-Rank the predicted value against the true value using two metrics.
1- Exact Match Rule: you compare the two strings after normalization and remove any special characters. report 1 if both values are literally and semantically equal and 0 otherwise.
2- Hierarchical/Categorical Match Rule: report 1 if the predicted value is under a subcategory or hierarchically belongs to the true value or is a synonym and 0 otherwise.
- Example:
List of pairs: [[music, art], [painter, artist],[football player, soccer player], [lawyer, judge], [lawyer, player]]
Answer: [[0,1],[0,1],[1,1],[0,1],[0,0]]
- Question:
-List of pairs: {**ListOfPairs**}
-Note: refine each pair and return Answer for exactly {**length(ListOfPairs)**} pairs without explanation.
-Finally: make sure you return only {**length(ListOfPairs)**} pair of answers.
Answer:

A.4 Full Details of Our GLOW-Bench

Table 5: Our GLOW-Bench benchmark characterizes each Open-World Question Template (OW-QT) across four KG-based features: (1) Knowledge Domain (KD): Domain-Specific (DS), Entertainment (E), or Generic (G); (2) Target Entity Type (ET); (3) Reasoning Hops (RH); and (4) Multiple Choice Count (MCC).

OW-QT	KG	KD	Target ET	RH	#Class	MCC	Label Type	Reasoning Path
1	BioKG	DS	Drug	1	2	2–4	Kingdom	Kingdom
2		DS	Drug	1	13	8–16	Superclass	Superclass
3		DS	Drug	2	29	16–32	Class	RelatedPubMed → Class
4		DS	Protein	1	18	16–32	SPECIES	SPECIES
5		DS	Protein	2	15	8–16	R.Keyword	RelatedPubMed → R.Keyword
6		DS	Protein	1	4	2–4	Family	Family
7	CrunchBase	DS	Investor	1	12	8–16	Country	Country
8		DS	Investor	1	14	8–16	InvestRegion	InvestRegion
9		DS	Investor	1	6	4–8	PositionTitle	PositionTitle
10		DS	Investor	1	11	8–16	Company	Company
11	LinkedMDB	E	Film	1	15	8–16	Language	Language
12		E	Film	1	27	16–32	Country	Country
13		E	Film	2	39	32+	Producer	Sequel → Producer
14		E	Film	2	7	4–8	Genre	Sequel → Genre
15		E	Film	1	28	16–32	Subject	Subject
16	YAGO4	G	CreativeW	2	13	8–16	ProdComp	byArtist → ProdComp
17		G	CreativeW	2	14	8–16	Publisher	isBasedOn → Publisher
18		G	CreativeW	1	25	16–32	PublishLang	Author
19		G	CreativeW	2	11	8–16	Genre	ProdComp → Genre
20		G	CreativeW	1	6	4–8	Country	Country
21		G	Person	1	3	2–4	GivenAward	GivenAward
22		G	Person	1	91	32+	Nationality	Nationality
23		G	Person	1	4	4–8	GraduateOfOrg	GraduateOfOrg
24		G	Person	1	102	32+	Occupation	Occupation
25		G	Person	1	8	4–8	SpokenLang	SpokenLang

A.5 The GNN Training/Inference Technical Details.

- For each question pattern, we train a node classification GNN using GraphSAINT with RGCN as GNN convolutional layer to support heterogeneous KG subgraphs. The initial node embeddings are iteratively aggregated with embeddings received from neighboring nodes connected to a specific relation until the embeddings of all nodes converge. The final embedding of a v_t, our main question entity, is obtained through two aggregations: an outer aggregation over each relation type and an inner aggregation over neighbouring nodes $\mathcal{RC}$ of a specific relation and defined by RGCN (Schlichtkrull and Thomas N. Kipf, 2018) as follows:

$$h_i^{(l+1)} = \sigma(\sum_{r \in R} \sum_{j \in N_i^r} \frac{1}{c_{i,r}} W_r^{(l)} h_j^{(l)} + W_0^{(l)} h_i^{(l)}) \quad (2)$$

  where l is an RGCN layer, $h_j^{(l+1)}$ is the hidden embedding of node j at layer $l+1$, σ is element-wise activation function, N_i^r denotes the set of neighbour indices of node i under relation $r \in \mathcal{R}$, $c_{i,r}$ is a normalization constant that can either be learned or chosen in advance (such as $c_{i,r} = |N_i^r|$), $W_r^{(l)}$ is the weight matrix for relation r at layer l, and W_0^l is the initial weight matrix at layer l.

- To construct the training set, we exclude the GLOW-Bench question entity nodes from the training set and extract 1-hop in and out neighbor nodes subgraphs using the KGTOSA sampler (Abdallah et al., 2024). This ensures scalability to large KGs and yields task-relevant and diverse subgraphs for effective training.

- GraphSAINT Training HyperParameters: We set the input dimension (D = 128), hidden channels dimension=64, and number of layers (L = 2) of our GNN module with dropout rate 0.5 applied to each layer. We train the model with the Adam optimizer, learning rate from 5e-4, 5e-3, 1e-3, 2e-3.
- Subgraph Sampling Hyperparameters: batch-size=20000, walk-length=2, num-steps=10.
- At inference time, the node classification model M is resolved using the KG name, the question's main entity type, and the target edge. The model M predicts the top-k answer classes based on the Log-Likelihood for each class and feeds them to the LLM prompt. The trained GNN model is hosted via an inference API. At runtime, this API returns the top-K predicted answer nodes, which are then passed to the GLOW pipeline as supportive evidence for the LLM's reasoning.

A.6 The OWA-QA Detailed Results:

Table 6: The detailed accuracy (%) of GLOW compared to baseline models on existing question answering datasets from (Hu et al., 2024) and our developed benchmark GLOW-Bench. All results below use Qwen3-8B as the underlying LLM. Best results are marked in **bold**.

LLM-Model	Dataset	RH	AskGNN	GraphSAINT	LLM-Only	GCR	**GLOW-L**	**GLOW-G**	**GLOW-N**	**GLOW-GN**
Qwen3:8b	Arxiv2023	1	62	41	31	4	59	64	66	**81**
	ogbn-arxiv	1	70	59	12	14	63	71	60	**77**
	ogbn-product	1	29	45	34	5	41	44	51	**57**
	drug-superclass	1	88	84	29	3	50	77	85	**92**
	drug-kingdom	1	98	96	88	94	91	**98**	97	96
	protein-SPECIES	1	75	89	8	16	19	71	91	**92**
	protein-FAMILY	1	67	70	9	13	35	39	69	**71**
	film-country	1	55	55	40	4	40	**60**	57	55
	film-subject	1	77	81	15	7	50	30	**82**	80
	film-language	1	**87**	83	70	13	73	80	85	86
	person-nationality	1	86	89	27	38	45	89	**91**	90
	parson-graduateOfOrg	1	51	53	3	25	27	36	54	**55**
	person-occupation	1	48	46	7	12	15	38	46	**53**
	person-spokenLang	1	86	83	70	61	83	**89**	83	85
	person-givenAward	1	93	91	7	50	75	93	93	**93**
	CWork-PublishedLang	1	75	80	50	60	55	55	85	**95**
	CWork-country	1	72	51	64	43	62	75	48	**79**
	person-title	1	79	72	5	41	63	47	77	**83**
	person-Company	1	97	95	3	93	30	21	96	**97**
	person-InvestementRegion	1	98	94	7	46	95	87	96	**100**
	person-InvestementCountry	1	**98**	95	38	14	94	55	97	97
	drug-class	2	46	68	11	15	33	45	70	**73**
	protein-keyword	2	35	42	5	7	33	53	44	**57**
	film-genre	2	54	51	50	25	60	**62**	51	52
	film-producer	2	**35**	21	7	7	14	24	21	28
	CWork-ProductionCompany	2	20	22	10	29	21	**31**	25	28
	CWork-Genere	2	20	22	16	7	22	**33**	19	27
	CWork-publisher	2	31	28	33	17	58	**62**	27	32

Table 7: The detailed accuracy (%) of GLOW compared to baseline models on existing question answering datasets from (Hu et al., 2024) and our developed benchmark GLOW-Bench. All results below use GPT-4o-Mini (OpenAI et al., 2024) as the underlying LLM. Best results are marked in **bold**.

LLM-Model	Dataset	RH	AskGNN	GraphSAINT	LLM-Only	**GLOW-L**	**GLOW-G**	**GLOW-N**	**GLOW-GN**
GPT-4o-Mini	Arxiv2023	1	N/A	41	35	58	59	48	**62**
	ogbn-arxiv	1	N/A	41	35	58	59	48	**67**
	ogbn-product	1	N/A	45	N/A	N/A	N/A	N/A	N/A
	drug-superclass	1	N/A	84	8	47	57	73	**87**
	drug-kingdom	1	N/A	96	31	**100**	**100**	98	99
	protein-SPECIES	1	N/A	89	6	10	77	92	**93**
	protein-FAMILY	1	N/A	70	11	53	43	69	**72**
	film-country	1	N/A	55	47	47	**71**	57	58
	film-subject	1	N/A	**81**	15	73	38	76	75
	film-language	1	N/A	83	80	80	80	86	**88**
	person-nationality	1	N/A	89	43	56	89	**92**	87
	parson-graduateOfOrg	1	N/A	53	19	22	33	55	**57**
	person-occupation	1	N/A	46	7	23	30	46	**52**
	person-spokenLang	1	N/A	83	70	81	**91**	83	89
	person-givenAward	1	N/A	91	6	68	**96**	93	91
	CWork-PublishedLanguage	1	N/A	80	65	60	65	83	**87**
	CWork-country	1	N/A	51	62	64	77	48	**83**
	person-title	1	N/A	72	5	38	55	74	**77**
	person-company	1	N/A	95	3	6	30	96	**98**
	person-InvestementRegion	1	N/A	94	9	7	96	**100**	96
	person-InvestementCountry	1	N/A	95	41	44	**100**	97	97
	drug-class	2	N/A	68	8	27	45	**69**	67
	protein-keyword	2	N/A	42	7	28	44	40	**45**
	film-genre	2	N/A	51	68	65	**70**	50	53
	film-producer	2	N/A	21	20	17	20	21	**31**
	CWork-ProductionCompany	2	N/A	22	10	28	**42**	25	32
	CWork-Genere	2	N/A	22	27	27	**72**	19	44
	CWork-publisher	2	N/A	28	62	70	**79**	25	75

Table 8: The detailed accuracy (%) of GLOW compared to baseline models on existing question answering datasets from (Hu et al., 2024) and our developed benchmark GLOW-Bench. All results below use DeepSeek-V3 (DeepSeek-AI, 2024) as the underlying LLM. Best results are marked in **bold**.

LLM-Model	Dataset	RH	AskGNN	GraphSAINT	LLM-Only	**GLOW-L**	**GLOW-G**	**GLOW-N**	**GLOW-GN**
DeepSeek-V3	Arxiv2023	1	N/A	41	17	50	57	61	**73**
	ogbn-arxiv	1	N/A	59	58	60	62	63	**65**
	ogbn-product	1	N/A	45	N/A	N/A	N/A	N/A	N/A
	drug-superclass	1	N/A	84	3	32	48	73	**86**
	drug-kingdom	1	N/A	96	9	88	**100**	98	96
	protein-SPECIES	1	N/A	89	2	8	79	92	**93**
	protein-FAMILY	1	N/A	70	21	76	**87**	69	81
	film-country	1	N/A	55	57	65	70	57	**71**
	film-subject	1	N/A	81	23	80	38	80	**83**
	film-language	1	N/A	83	81	86	86	86	**93**
	person-nationality	1	N/A	89	72	81	87	91	**93**
	parson-graduateOfOrg	1	N/A	53	32	48	43	52	**54**
	person-occupation	1	N/A	46	19	34	34	46	**53**
	person-spokenLang	1	N/A	83	79	79	82	83	**85**
	person-givenAward	1	N/A	91	3	56	58	93	**94**
	CWork-PublishedLanguage	1	N/A	80	47	50	50	84	**90**
	CWork-country	1	N/A	51	68	75	**87**	48	79
	person-title	1	N/A	72	41	41	63	74	**77**
	person-company	1	N/A	95	3	24	33	96	**97**
	person-InvestementRegion	1	N/A	94	7	21	100	89	**100**
	person-InvestementCountry	1	N/A	95	44	58	100	97	**100**
	drug-class	2	N/A	68	7	21	39	69	**70**
	protein-keyword	2	N/A	42	3	40	**59**	49	52
	film-genre	2	N/A	51	53	60	58	50	**62**
	film-producer	2	N/A	21	10	30	**50**	14	34
	CWork-ProductionCompany	2	N/A	22	28	53	**57**	25	53
	CWork-Genere	2	N/A	22	18	22	27	16	**28**
	CWork-publisher	2	N/A	28	70	79	**87**	16	70

Table 9: The detailed accuracy (%) of GLOW compared to baseline models on existing question answering datasets from (Hu et al., 2024) and our developed benchmark GLOW-Bench. All results below use DeepSeek-R1-Distill-Qwen:7B (DeepSeek-AI, 2025) as the underlying LLM. Best results are marked in **bold**.

LLM-Model	Dataset	RH	AskGNN	GraphSAINT	LLM-Only	GCR	**GLOW-L**	**GLOW-G**	**GLOW-N**	**GLOW-GN**
	Arxiv2023	1	48	41	24	7	44	55	59	**67**
	ogbn-arxiv	1	50	59	13	15	30	30	61	**63**
	ogbn-product	1	19	45	27	12	36	38	43	**51**
	drug-superclass	1	53	84	20	5	20	21	**69**	68
	drug-kingdom	1	98	96	60	90	90	**100**	98	**100**
	protein-SPECIES	1	71	89	1	12	4	62	88	**90**
	protein-FAMILY	1	62	70	7	23	30	35	68	**69**
	film-country	1	56	55	17	34	44	52	57	**64**
	film-subject	1	78	81	8	8	11	34	76	**82**
	film-language	1	86	83	73	32	73	66	86	**88**
	person-nationality	1	85	89	22	41	30	79	**91**	89
	parson-graduateOfOrg	1	50	53	5	17	19	36	53	**54**
	person-occupation	1	47	46	3	12	15	31	46	**54**
DeepSeek-R1-Distill-Qwen:7B	person-spokenLang	1	82	83	35	56	41	79	81	**83**
	person-givenAward	1	89	91	8	23	50	70	90	**92**
	CWork-PublishedLanguage	1	71	80	50	44	50	55	77	**81**
	CWork-country	1	62	51	37	42	41	**70**	48	66
	person-title	1	73	72	5	51	44	52	80	**82**
	person-company	1	93	95	3	90	30	19	**96**	92
	person-InvestementRegion	1	92	94	3	33	64	96	**100**	**100**
	person-InvestementCountry	1	93	95	11	56	72	94	97	**97**
	drug-class	2	41	68	4	13	15	24	69	**71**
	protein-keyword	2	31	42	3	8	18	22	32	**38**
	film-genre	2	53	51	35	18	50	56	50	**57**
	film-producer	2	30	21	7	9	7	7	14	**24**
	CWork-ProductionCompany	2	18	22	10	23	17	21	25	**32**
	CWork-Genere	2	17	22	11	17	11	16	16	**27**
	CWork-publisher	2	25	28	12	23	20	**40**	16	26

Table 10: The detailed accuracy (%) of GLOW compared to baseline models on existing question answering datasets from (Hu et al., 2024) and our developed benchmark GLOW-Bench. All results below use IBM Granite-3.3-8B-instruct (Saon et al., 2025) as the underlying LLM. Best results are marked in **bold**.

LLM-Model	Dataset	RH	AskGNN	GraphSAINT	LLM-Only	GCR	**GLOW-G**	**GLOW-N**	**GLOW-GN**
	Arxiv2	1	63	41	14	14	32	63	**77**
	ogbn-arxiv	1	66	59	5	21	**31**	55	**73**
	ogbn-product	1	30	45	22	13	45	52	**54**
	drug-superclass	1	86	84	0	9	22	73	**71**
	drug-kingdom	1	97	96	8	98	95	98	**99**
	protein-SPECIES	1	75	89	2	18	45	93	**90**
	protein-FAMILY	1	67	70	0	33	36	69	**72**
	film-country	1	56	56	48	60	**66**	58	55
	film-subject	1	76	81	12	11	27	81	**85**
	film-language	1	84	83	87	46	80	87	**88**
	person-nationality	1	86	89	52	88	79	91	**92**
	parson-graduateOfOrg	1	52	53	6	14	31	56	**57**
	person-occupation	1	47	46	8	23	23	46	**50**
Granite-3.3:8B-Instruct	person-spokenLang	1	85	83	48	75	52	83	**88**
	person-givenAward	1	93	92	3	25	63	91	**94**
	CWork-PublishedLanguage	1	81	80	45	35	80	81	**94**
	CWork-country	1	72	51	58	69	**75**	59	69
	person-title	1	78	72	14	55	64	80	**83**
	person-company	1	97	96	3	96	42	96	**97**
	person-InvestementRegion	1	99	95	4	29	82	98	**100**
	person-InvestementCountry	1	**99**	96	39	69	78	96	97
	drug-class	2	47	68	0	18	27	70	**70**
	protein-keyword	2	35	42	2	10	36	9	**38**
	film-genre	2	54	51	35	15	52	54	**57**
	film-producer	2	**35**	21	5	5	30	23	29
	CWork-ProductionCompany	2	22	22	7	18	21	25	**32**
	CWork-Genere	2	21	22	16	22	**50**	17	33
	CWork-publisher	2	32.7	28	33	46	**58**	27	36